

Finite-time Analysis of the Multi-armed Bandit Problem

by Auer, Cesa-Bianchi & Fischer (2002), *Machine Learning* 47:235–256

Paper Summary – Seminar “Decision Making Algorithms”

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1 Motivation and Problem

How can an agent maximize cumulative reward when making repeated decisions under uncertainty? Its canonical form is the *multi-armed bandit*: a player faces K machines (“arms”) with unknown reward distributions and must choose one each round, trading off *exploitation* (playing the best-looking arm) against *exploration* (trying others). Lai and Robbins (1985) showed regret must grow at least logarithmically in the number of plays n , but only *asymptotically*. This paper gives efficient policies achieving logarithmic regret *uniformly over time* – matching the finite horizon of real-world applications.

2 Setup

Each arm i yields independent identically distributed rewards with unknown mean μ_i , independent across arms. Let $T_i(n)$ be the number of plays of arm i in the first n rounds, $\bar{x}_i$ its sample mean, $\mu^* = \max_i \mu_i$ the best mean, and $\Delta_i = \mu^* - \mu_i$ the suboptimal gap. The performance of the policies is measured with regret, the difference between the expected reward of playing the policy and playing always the best arm:

$$\text{Regret} = \mu^* n - \sum_{j=1}^K \mu_j \mathbb{E}[T_j(n)]$$

3 Algorithms, Guarantees and Proof Ideas

3.1 UCB1

At round n , play the machine j that maximizes $\bar{x}_j + \sqrt{(2 \ln n)/n_j}$, where $\bar{x}_j$ is the average reward of machine j and n_j is its number of plays so far. Initialization requires playing each of the K machines once. This mechanism allows machines with a low average reward to be played again, ensuring that a high-performing machine that initially performed poorly due to bad luck can still be discovered.

3.1.1 Theorem 1

For all $K > 1$, the expected cumulative regret is bounded logarithmically relative to the number of plays n . This regret bound is derived using the Chernoff-Hoeffding bound. The main objective of the proof is to bound the probability of playing a suboptimal machine that does not maximize the true mean reward.

3.2 UCB1-Tuned

By changing the exploration bonus of UCB1, we can create a variant called UCB1-Tuned. This version shows better results in experiments, but a mathematical regret bound cannot be proven. The new exploration bonus is defined as $\sqrt{(\ln n/n_j) \min\{1/4, V_j(n_j)\}}$, where

$$V_j(s) = \left(\frac{1}{s} \sum_{\tau=1}^s X_{j,\tau}^2 \right) - \bar{x}_j^2 + \sqrt{(2 \ln n)/s}.$$

3.3 UCB2

To minimize the large constant factors of UCB1, UCB2 introduces an epoch-based structure. Instead of playing a selected machine only once, the algorithm plays the chosen machine j sequentially for a specific block of trials, called an epoch, of length $\tau(r_j + 1) - \tau(r_j)$, where $\tau(r) = \lceil (1 + \alpha)^r \rceil$ and $0 < \alpha < 1$. At the end of these epochs, an arm j is selected by maximizing the sum of its average reward $\bar{x}_j$ and the exploration bonus term

$$a_{n,r_j} = \sqrt{\frac{(1 + \alpha) \ln(en/\tau(r_j))}{2\tau(r_j)}}.$$

3.3.1 Theorem 2

For all $\alpha > 0$ and $K > 1$, the expected regret is bounded logarithmically. Similar to UCB1, the proof utilizes the Chernoff-Hoeffding bound to control suboptimal plays. However, it splits the analysis to treat early epoch errors as a fixed penalty, which introduces a time-independent constant term c_α/Δ_i into the cumulative regret equation; a smaller α optimizes the long-term rate but causes this constant to increase.

3.4 ε_n -GREEDY

With probability $\varepsilon_n = \min\{1, cK/(d^2n)\}$ play a random arm, else the empirical best. Here $c > 0$ a tuning constant, K the number of arms, d a known lower bound on the gap between the best and second best arm. A decaying ε_n is essential since constant ε gives linear regret.

3.4.1 Theorem 3

For $n \geq cK/d$, it bounds $P\{\text{pull suboptimal arm } j \text{ at round } n\}$ by order $c/(d^2n) + o(1/n)$, which integrates out to logarithmic regret, given a known lower bound $d \leq \min_i \Delta_i$. The proof splits that probability into a random exploration term

and an exploitation term. The exploration term is already an exact quantity – there is nothing to bound – so all the risk is in the exploitation term, which is bounded via Chernoff-Hoeffding and re-expressed using only j 's purely random pulls, whose count is controlled directly by Bernstein's inequality.

3.5 UCB1-Normal

Unlike the other policies, which assume bounded rewards $[0, 1]$, UCB1-Normal assumes normal rewards with both mean and variance unknown, estimating each arm's variance from its data gathered through its trials. For each round $n = 1, 2, \dots$:

- If there is a machine played less than $\lceil 8 \log n \rceil$ times play this machine.
- Otherwise play machine j that maximizes

$$\bar{x}_j + \sqrt{16 \frac{q_j - n_j \bar{x}_j^2}{n_j - 1} \cdot \frac{\ln(n-1)}{n_j}},$$

where q_j is the sum of squared rewards obtained from machine j .

- Update $\bar{x}_j$ and q_j with obtained reward x_j .

3.5.1 Theorem 4

For all $K > 1$, if UCB1-Normal is run on K arms with normal distributions $P_1, \dots, P_K$ with means μ_i , variances σ_i^2 , its expected regret after n plays is at most:

$$256(\log n) \left(\sum_{i: \mu_i < \mu^*} \frac{\sigma_i^2}{\Delta_i} \right) + \left(1 + \frac{\pi^2}{2} + 8 \log n \right) \left(\sum_{j=1}^K \Delta_j \right).$$

Each suboptimal arm contributes $256 \sigma_i^2 / \Delta_i$, and the forced exploration shows the regret in the $\lceil 8 \log n \rceil$ term. This results in a higher logarithmic factor than the other policies.

4 Experiments

Our simulations reproduce all policies across distributions of varying difficulties, comparing UCB1-Tuned, UCB2, and ε_n -GREEDY (at several c values) on Distribution 2 $[0.9, 0.8]$, Distribution 12 $[0.9, 0.8 \times 3, 0.7 \times 3, 0.6 \times 3]$, Distribution 13 $[0.9, 0.8 \times 9]$, and Distribution 14 $[0.55, 0.45 \times 9]$. Figure 1 reports the share of plays on the best arm and Figure 2 reports cumulative regret.

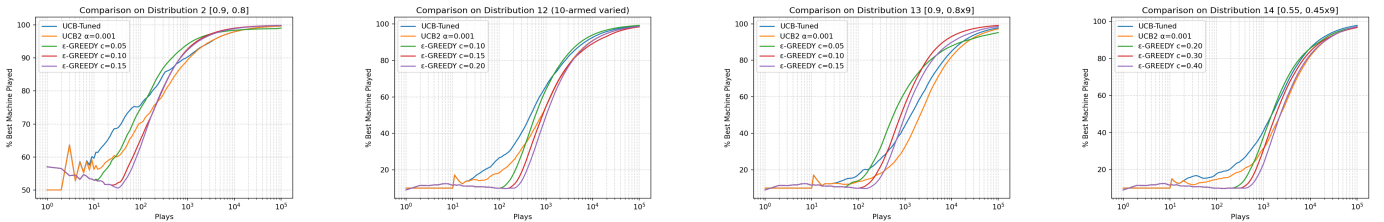


Figure 1: Cross-algorithm comparison – % best machine played across the four distributions.

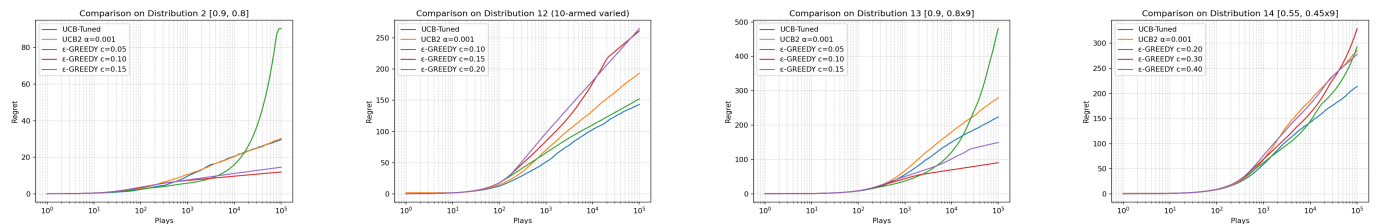


Figure 2: Cross-algorithm comparison – cumulative regret across the four distributions.

On easy two-arm problems (Dist 2) a well-tuned ε_n -GREEDY ($c \approx 0.10$) is the clear winner, with UCB1-Tuned and UCB2 close behind. But ε_n -GREEDY is fragile: on Dist 12 and 14, where uniform exploration wastes equal pulls on both terrible and competitive arms, it breaks down while UCB1-Tuned adapts and dominates – the paper's signature finding. Dist 13 shows the weakness is specific to widely spread losers rather than arm count, since ε_n -GREEDY again wins there. The best c spans roughly an $8\times$ range across distributions, confirming the authors' remark that no single value works well for all problems. Overall UCB2 tracks UCB1-Tuned but stays consistently slightly worse, while UCB1-Tuned is stable and largely insensitive to reward variance.

5 Strengths, Weaknesses and Takeaway

The paper's primary strength is being the first to prove logarithmic regret **uniformly over finite n** for any bounded-support reward distribution, with Theorems 1–3 remaining valid under mild reward dependence. Its limitations are equally concrete: UCB1's leading constant is about $16\times$ worse than the Lai-Robbins optimum; UCB1-NORMAL rests on two conjectures verified only numerically; ε_n -GREEDY requires prior knowledge of d and tuning of c ; and the empirical winner UCB1-Tuned has no theoretical proof. Despite these gaps, simple well-analysed index policies make near-optimal bandit performance practical, and UCB-style optimism laid the groundwork in later bandit and reinforcement learning research.